Lab #4: SLAM and Nav2 (ROS 2 Humble)

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I. SLAM and Nav2 with `my_robot`

This lab guides you through simulation, mapping (SLAM Toolbox), and localization/navigation (Nav2) using `my_robot`.
The following prerequisites are required:

- **ROS 2** Humble installed and sourced
- Gazebo and RViz2 installed
- `my_robot` built and sourced

```
colcon build && source install/setup.bash
```

- Robot publishes `/robot_description` and a laser `/scan`

A. Simulation + RViz

Displays the robot in Gazebo and publishes `/robot_description` and `/scan` for RViz.

```
ros2 launch my_robot launch_sim.launch.py
```

In RViz:

- Set Fixed Frame (e.g., `odom`)
- Add `RobotModel` (uses `/robot_description`)
- Add `LaserScan` on `/scan`

B. Mapping with SLAM Toolbox

Start online SLAM and drive the robot to cover the space.

```
ros2 launch my_robot launch_slam.launch.py
```

Tip

- Teleop: `ros2 run teleop_twist_keyboard teleop_twist_keyboard`
- Check topics/frames: `ros2 topic list`, `ros2 run tf2_tools view_frames`
- View map in RViz: add `Map` (topic `/map`)

Save the map:

```
ros2 run nav2_map_server map_saver_cli -f ~/maps/my_world
```

C. Localization and Navigation (Nav2)

Launch Nav2 (AMCL + planners + controllers) and navigate in RViz.

```
ros2 launch my_robot launch_nav.launch.py
```

The overall workflow is as follows:

- In RViz set `Fixed Frame` to `map`
- Use `2D Pose Estimate` to set initial pose
- Use `Nav Goal` to send a goal

D. Summary

```
# Simulation
ros2 launch my_robot launch_sim.launch.py

# Mapping
ros2 launch my_robot launch_slam.launch.py

# Localization + Navigation
ros2 launch my_robot launch_nav.launch.py
```